

# EZvest API Costing & Future Cost Plan

Prepared 14 May 2026. This document estimates Claude API usage, payment fees, backend costs, and scale-up risk for the current EZvest app. Amounts use ZAR where useful and assume USD/ZAR = 16.4, matching the app's current backend estimate.

Important: this is an operating-cost model, not accounting or tax advice. Recheck provider pricing before launch or fundraising, because API and platform prices can change.

## 1. Current App Cost Model

- Frontend: static HTML/CSS/JavaScript, currently suited to GitHub Pages or another static host.
- Backend: Supabase Auth, Postgres, Edge Functions, rate-limit tables, function event logging, and a shared market cache.
- AI: Supabase Edge Function named finbot calls Anthropic Claude Messages API.
- Current default model in code/docs: claude-sonnet-4-5 unless ANTHROPIC\_MODEL is changed.
- Billing: Paystack recurring plan checkout and webhook sync.
- Paid plan credits in the repo: Basic 50, Pro 100, Enterprise 300 monthly FinBot credits.
- Credit costs in code: chat = 1 credit, news = 2 credits, full FinBot analysis = 5 credits.

### Current paid plans in repo

| Plan       | Price/mo | Credits | Full analyses if all credits used |
|------------|----------|---------|-----------------------------------|
| Basic      | R199.00  | 50      | 10                                |
| Pro        | R399.00  | 100     | 20                                |
| Enterprise | R799.00  | 300     | 60                                |

## 2. Claude API Pricing Used

Official Anthropic API pricing is per million tokens \(\MTok\). Input tokens and output tokens are priced separately. Your current code also records token usage and estimates cost after each successful FinBot response.

### Claude API list prices used in this report

| Model                   | Input / MTok | Output / MTok | Recommended EZvest use                                |
|-------------------------|--------------|---------------|-------------------------------------------------------|
| Claude Haiku 4.5        | \$1          | \$5           | Low-cost chat, summaries, simple education flows      |
| Claude Sonnet 4.5 / 4.6 | \$3          | \$15          | Current default; best balanced model for FinBot       |
| Claude Opus 4.6         | \$5          | \$25          | Premium future tier or complex portfolio/risk reports |

Anthropic also lists 50% discounted Batch API pricing. That is useful for future background jobs, bulk report regeneration, or scheduled portfolio reviews, but not for live user chat because batch work is asynchronous.

## 3. Estimated Claude Cost per User Action

The estimates below use Sonnet pricing because Sonnet is your current default and likely the best production baseline. Real cost depends on prompt size, chat history length, market data included in the prompt, and output length.

**Typical request economics - Claude Sonnet**

| Request              | Credits | Tokens in/out | Claude cost      | Cost / credit | Note                                      |
|----------------------|---------|---------------|------------------|---------------|-------------------------------------------|
| Chat                 | 1       | 1200/500      | \$0.0111 / R0.18 | R0.18         | Typical short FinBot chat                 |
| News analysis        | 2       | 2500/1200     | \$0.0255 / R0.42 | R0.21         | Article/news explanation                  |
| Full FinBot analysis | 5       | 3800/2600     | \$0.0504 / R0.83 | R0.17         | Average screener/DCF/risk/rep request     |
| Heavy builder/risk   | 5       | 4200/4200     | \$0.0756 / R1.24 | R0.25         | Upper practical scenario near report caps |

**Upper practical request cases - Claude Sonnet**

| Case              | Tokens in/out | Credits | Cost             | Cost / credit |
|-------------------|---------------|---------|------------------|---------------|
| Chat with history | 11400/900     | 1       | \$0.0477 / R0.78 | R0.78         |
| News max cap      | 3200/1800     | 2       | \$0.0366 / R0.60 | R0.30         |
| Builder max cap   | 3600/4200     | 5       | \$0.0738 / R1.21 | R0.24         |

**4. AI Cost If Users Spend All Credits**

This is the stress test most relevant to pricing. It assumes every paid user spends every monthly credit. In practice, many subscribers will not use 100% of credits, which improves margin.

**Monthly Claude cost per subscriber if all credits are used**

| Plan       | Included credits | All as avg full reports | All as heavy reports | AI margin before other costs |
|------------|------------------|-------------------------|----------------------|------------------------------|
| Basic      | 50               | R8.27                   | R12.40               | 93.8%                        |
| Pro        | 100              | R16.53                  | R24.80               | 93.8%                        |
| Enterprise | 300              | R49.59                  | R74.39               | 90.7%                        |

**5. Paystack Fees and Net Revenue**

Paystack South Africa lists local card pricing at 2.9% + R1 excluding VAT. This report assumes EZvest absorbs the fee and adds 15% VAT on the Paystack fee. If you pass fees to customers, net revenue improves.

**Approximate Paystack impact by plan**

| Plan       | Customer pays | Paystack fee incl VAT | Net before AI/backend | Net after heavy AI use |
|------------|---------------|-----------------------|-----------------------|------------------------|
| Basic      | R199.00       | R7.79                 | R191.21               | R178.81                |
| Pro        | R399.00       | R14.46                | R384.54               | R359.75                |
| Enterprise | R799.00       | R27.80                | R771.20               | R696.81                |

**6. Supabase, Hosting, and Backend Costs**

- GitHub Pages: official GitHub docs state Pages is available for public repositories on GitHub Free. For the current static frontend, hosting can be R0/month if the repository is public and GitHub Pages remains acceptable for commercial use constraints.
- Supabase Free: useful for testing, but free projects pause after inactivity and have lower included quotas. Not recommended as the long-term production plan.
- Supabase Pro: current public pricing shows \$25/month base with the first Micro compute covered by included compute credits, 100k monthly active users, 8 GB disk, 250 GB egress, 100 GB storage, and 2M Edge Function invocations. At USD/ZAR 16.4 this is about R410/month before VAT/exchange fees.
- Supabase Edge Function overage: official docs list Pro overage at \$2 per additional 1M invocations after the included 2M. EZvest should remain well below this for a long time if market-data caching stays enabled.
- Custom domain through Supabase is listed at \$10/month, but if you use GitHub Pages/custom DNS directly this may not be needed.
- Market data: current code mostly uses public RSS/news feeds and Yahoo-style quote endpoints through your own Edge Function. Future premium market data vendors can become a major cost if you add real-time exchange-grade prices.

## 7. Launch Budget Recommendations

### Recommended monthly budget by launch phase

| Phase         | Expected usage          | Expected fixed costs | Variable AI/payment costs | Comment                                                          |
|---------------|-------------------------|----------------------|---------------------------|------------------------------------------------------------------|
| Pre-launch    | Internal testing        | R0-R410              | R0-R200                   | Free/Supabase Pro dependent on needs                             |
| Soft launch   | 0-100 paid users        | About R410           | R100-R2,000               | Claude costs stay small; pay more                                |
| Early revenue | 100-1,000 paid users    | R410-R1,500          | R1k-R20k                  | Monitor token usage and costs                                    |
| Growth        | 1,000-10,000 paid users | R1.5k-R10k+          | R20k-R200k+               | Consider Haiku routing, batch data vendors, and stronger caching |

## 8. Future Cost Risks

- Long chat histories: chat costs can rise sharply if every message includes large prior history. Keep summarizing old conversation context.
- Large report outputs: builder/risk/DCF modes have the highest max\_tokens caps. Add per-mode cost telemetry dashboards before scaling.
- Model upgrades: Opus is materially more expensive than Sonnet. Reserve Opus for an Enterprise-only feature or one-off premium reports.
- Market data licensing: public feeds are cheap but limited. Real-time stock APIs, exchange data, or South African market data can add fixed monthly vendor costs.
- Abuse and scraping: even rejected Edge Function calls can count as invocations. Add frontend throttles, backend rate limits, and possibly WAF/CDN rules before public marketing.
- Support and refunds: payment fees, chargebacks, and customer support load are real costs even when AI cost is low.

## 9. Practical Cost Controls to Add Next

- Route simple chat and article summaries to Haiku 4.5; keep Sonnet for full analyses. This can cut lightweight AI cost by roughly two-thirds.

- Keep Sonnet as the default for DCF, risk, builder, and technical reports until quality data says Haiku is good enough.
- Set hard monthly Anthropic spend alerts at R500, R1,500, R5,000, and R10,000 during launch.
- Store actual anthropic\_usage rows or daily aggregates from the existing usageCost object returned by finbot.
- Add a daily dashboard: requests, tokens, Claude cost, credits spent, revenue by tier, AI cost per paid user, and gross margin.
- Consider lowering chat history sent to Claude from 8 turns to 4 turns plus a running summary.
- Use Batch API for non-interactive future jobs such as weekly portfolio summaries; official pricing lists a 50% discount.

## 10. Bottom Line

With the current credit design, Claude API cost is unlikely to be the main problem at launch. Payment fees and fixed backend reliability costs are more visible early. Even if a Basic user spends every credit on heavy reports, estimated Claude cost is roughly R12.40/month against a R199 subscription before payment fees. The bigger future risks are model choice, long chat histories, abuse, and premium market data vendors.

Suggested launch setup: Supabase Pro, GitHub Pages or equivalent static hosting, Claude Sonnet default for full reports, Haiku for future lightweight chat/news, Paystack fees either absorbed in pricing or transparently passed through.

## Sources

- Anthropic Claude API pricing: <https://platform.claude.com/docs/en/about-claude/pricing>
- Supabase pricing: <https://supabase.com/pricing>
- Supabase Edge Function pricing: <https://supabase.com/docs/guides/functions/pricing/>
- Paystack South Africa pricing: <https://paystack.com/za/pricing>
- Paystack transaction fee article: <https://support.paystack.com/en/articles/2130306>
- GitHub Pages availability: <https://docs.github.com/en/pages/getting-started-with-github-pages/github-pages-limits>
- Local app files reviewed: `supabase/functions/finbot/index.ts`, `supabase/functions/market-data/index.ts`, `README.md`